

Visual impact of two wind farm sites on recreation areas

Gronau, North Rhine-Westphalia · Visibility analysis in ArcGIS Pro (Spatial Analyst)

Carlo Piccillo · Course exercise Tü 3 "Windparkplanung", "GIS and Geodata Expert" programme, GIS-Akademie Berlin · 2026

The task, its parameters and the scenario data were set by the course. The data checks, the analysis, the map and this assessment are my own work.

1 Question

Two sites are under discussion for a wind farm of four turbines near Gronau. Which one affects the view from sports, leisure and recreation areas less? The course sets the threshold: the turbines are 240 m tall, a view counts as an impairment once 10 % of a turbine is visible, and people see from 1.6 m above the ground. In operational terms, an area is affected where at least the top 24 m of a turbine can be seen.

2 Data and source

Dataset	Content	Source
Surface model (DOM)	Buildings and vegetation included; 1 m cells, heights in m, no vertical reference	Course data; format matches the Geobasis NRW DOM1 (inferred, no metadata supplied)
Basis-DLM, settlement land use	1,232 polygons, of which 53 sports, leisure and recreation areas (125.6 ha) were used	Geobasis NRW (object IDs "DENW..."), dl-de/zero-2-0, via course data
Wind turbines	8 turbines at two sites (4 + 4), 150 m polylines with offsets of 216 m and 1.6 m	GIS-Akademie Berlin (exercise scenario)

The study area is the extent of the surface model. It covers the municipality of Gronau (Westf.), including Epe, at the Dutch border, within a 12.1 × 14.4 km bounding box.

3 Method

- **Threshold.** The 10 % rule is built into the turbine data: an observer offset of 216 m (90 % of 240 m) and a target offset of 1.6 m. If a sightline from 1.6 m above the surface reaches 216 m above a turbine base, the top 24 m of that turbine are visible. Sightlines are symmetric, so the turbines act as observers and every cell as a potential viewer.
- **Observers.** Each turbine is a straight 150 m line with two vertices. All eight have the same length and orientation (NNW–SSE), plausibly the rotor diameter seen from above. The tool uses line vertices as observers, which gives eight observer points per site; the output maximum of 8 confirmed this.
- **Viewshed.** Geodesic Viewshed with all sightlines, frequency output, refraction 0.13 and no distance limit; turbine base heights from the surface model; a 2 m grid snapped to the model, with the model's extent. Cells that no observer sees are NoData.
- **Recreation areas.** Basis-DLM features with OBJART 41008 (sports, leisure and recreation areas). Cemeteries form a separate class and are excluded.
- **Visible area.** Zonal Statistics as Table per area, with NoData ignored: visible cells × 4 m² give the area with a view of the turbines. The zone field is the Basis-DLM object ID, so each value traces back to the official object.

Checks: both outputs reach the maximum of 8 observers, and AREA = COUNT × 4. The viewsheds clipped to the recreation areas contain exactly the cell totals of the zonal statistics (103,918 and 169,237). A tool warning (010451) said the heights are not ellipsoidal; this amounts to a near-uniform vertical shift of the whole model, which leaves the relative geometry, and so the visibility, practically unchanged.

4 Result

Site	Area with a view	Share of 125.6 ha	Areas affected
Wind farm 1 (south-west)	41.6 ha	33 %	52 of 53
Wind farm 2 (north-east)	67.7 ha	54 %	53 of 53

Wind farm 1 has the smaller impact. Its turbines are visible from 26 ha (39 %) less recreation area than those of Wind farm 2.

Two findings qualify this. First, counting areas does not separate the sites: 52 of the 53 areas see at least a turbine tip of both. Second, over the whole study area the two sites are almost equally visible, from about 46 and 49 km². The difference comes from location, not from general visibility: several recreation areas close to Wind farm 2, at the northern edge of Gronau, are visible almost in full.

5 Analytical limits

- **The perimeter decides what "less impact" means.** Only recreation areas of the Basis-DLM settlement domain inside the surface model count. Woodland, open countryside and paths are not in the layer, although much of the recreation in the Münsterland happens there. The model also stops at the Dutch border, right next to Wind farm 1, so its visibility towards the Netherlands is not counted. The ranking holds within the exercise's perimeter, not in absolute terms.
- **Visible is not the same as disturbing.** The threshold is binary: a 24 m blade tip 10 km away counts as much as a full turbine at 1 km. Distance, the number of turbines in view, rotor movement and night lighting are not weighted. The viewshed does count up to 8 observers, so a graded reading is possible.
- **A surface model is not the ground.** It includes roofs and tree crowns, so where they exist the viewers stand 1.6 m above them. In a wooded park the model therefore overstates what people on the paths can see. Resampling from 1 m to 2 m (nearest neighbour) also keeps or drops narrow obstacles, such as single trees and hedges, almost at random.
- **The turbine geometry is a simplification.** The two observers of each turbine sit 75 m either side of the tower, at 216 m. If the line stands for the rotor, real blade tips at that distance from the tower reach only about hub height. The rotor orientation is also fixed.
- **The recreation class is mixed.** It includes allotment gardens and weekend or holiday-home areas, which are closer to housing than to leisure. The area-weighted total is dominated by a few large parks: per hectare, a sports ground used daily weighs no more than a rarely used green strip.
- **Not modelled:** noise, shadow flicker, legal distance rules and actual visitor numbers. Visibility is one criterion of a siting decision, not the decision.